Theory note S3a — Noticing a partial-band jammer: an energy test on pilots and the number of hops it needs

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Purpose. S0–S2 treat the jammer as a known input. S3a asks how many hops a receiver must observe before it can say that a partial-band jammer is present. The detector is a classical energy test, so every performance number has a closed form against which the simulator is checked. Mathematics used: Gaussian likelihoods, the Neyman–Pearson lemma, chi-square and binomial distributions.

1 The per-hop statistic

Each hop carries n known BPSK pilot symbols with amplitude $\sqrt{E_b} = 1$, received as $r_i = s_i + w_i$ exactly as in S0. The receiver subtracts the known pilots, which leaves noise only. As in S1, a hop lands in the jammed part of the band with probability ρ; there the noise density is $N_0 + N_J/\rho$ instead of N_0. With $\sigma^2 = N_0/2$ the per-hop statistic is

$$T = \frac{1}{\sigma^2}\sum_{i=1}^{n}(r_i - s_i)^2, \qquad \text{unjammed: } T \sim \chi^2_n, \qquad \text{jammed: } T/a \sim \chi^2_n, \tag{1}$$

$$a = 1 + \frac{N_J/N_0}{\rho}, \qquad \frac{N_J}{N_0} = \frac{E_b/N_0}{E_b/N_J}, \qquad E_b/N_J\,[\mathrm{dB}] = G_p - J/S\,[\mathrm{dB}], \tag{2}$$

where the last two relations are the S1 link budget. The noise ratio a is the only quantity through which J/S and ρ enter S3a. Energy statistics of this kind are the classical energy detector: with noise only, the sum of squared Gaussian samples follows a central chi-square distribution [1]. Here both hypotheses are central chi-square with different scales, because the jammer is modelled as noise.

At the S1 outage point ($J/S = 0.46\,\mathrm{dB}$) the noise ratio is $a = 12.7$ for the jammer's worst-case $\rho = 0.0095$, $a = 2.11$ for $\rho = 0.1$ and $a = 1.11$ for $\rho = 1$. A jammer that concentrates its power makes each jammed hop obvious, but few hops are jammed.

2 Why the energy test is optimal for one hop

For the n independent residual samples $x_i = r_i - s_i$, the likelihood ratio of "jammed" ($N(0, a\sigma^2)$) to "unjammed" ($N(0, \sigma^2)$) is

$$\Lambda(T) = \prod_{i=1}^{n} \frac{(2\pi a\sigma^2)^{-1/2}\, e^{-x_i^2/(2a\sigma^2)}}{(2\pi\sigma^2)^{-1/2}\, e^{-x_i^2/(2\sigma^2)}} = a^{-n/2}\exp\Big(\frac{T}{2}\big(1 - \tfrac{1}{a}\big)\Big). \tag{3}$$

For $a > 1$ it increases with T, so "$\Lambda > \eta$" is the same test as "$T > \gamma$". By the Neyman–Pearson lemma this test has the largest detection probability of all tests with the same false-alarm probability, whatever the value of a. Its performance follows from Eq. (1):

$$P_{\mathrm{fa}} = P(\chi^2_n > \gamma), \qquad P_{\mathrm{d}} = P(\chi^2_n > \gamma/a). \tag{4}$$

The receiver is assumed to know N_0 (a calibrated noise floor) and the pilots (perfect channel knowledge, as for coherent BPSK in S0). If the pilots were not subtracted, the signal energy would enter T and Eq. (3) would no longer reduce to an energy comparison.

3 A window of M hops: the count detector

The receiver flags each hop with $T > \gamma$ and declares "jammer present" when at least k of M hops are flagged. Hops are independent, so the count is binomial:

$$\text{no jammer: } \mathrm{Bin}(M, P_{\mathrm{fa}}), \qquad \text{jammer: } \mathrm{Bin}(M, p_1), \quad p_1 = \rho\, P_{\mathrm{d}} + (1-\rho)\, P_{\mathrm{fa}}. \tag{5}$$

For a given M and γ, k is the smallest count that keeps the window false-alarm probability at or below 10^{-3}. M^* is the smallest M whose window detection probability reaches 0.9, taking the best per-hop threshold from a grid of 70 values of P_{fa} between 10^{-9} and 0.5. Both targets are assumed values.

The count detector is simple but not optimal when $\rho < 1$: the optimal window test would add up the per-hop mixture log-likelihood ratios $\sum_j \ln\big(1-\rho+\rho\,\Lambda(T_j)\big)$, which requires knowing ρ and a. M^* is therefore what *this* detector needs; an optimal detector needs at most as many hops.

4 Two limits used as checks

Strong jammer (verification 3). If every jammed hop is flagged ($P_{\mathrm{d}} \to 1$) and false alarms are negligible ($P_{\mathrm{fa}} \to 0$, so $k = 1$), the window only has to contain one jammed hop:

$$1 - (1-\rho)^M \geq 0.9 \quad \Longleftrightarrow \quad M \geq \frac{\ln 0.1}{\ln(1-\rho)} \approx \frac{2.30}{\rho}. \tag{6}$$

At $J/S = 30\,\mathrm{dB}$ the computed M^* must be within one hop of the ceiling of this bound. It can be one hop *below* it because the allowed false alarms ($P_{\mathrm{fa}} \lesssim 10^{-3}/M$ per hop) add slightly to p_1.

Full-band jammer (verification 4). At $\rho = 1$ every hop is jammed and all Mn residual samples are i.i.d.; the likelihood ratio of the whole window has the form of Eq. (3) with the pooled energy $\sum_j T_j \sim \chi^2_{Mn}$ (scaled by a when jammed). The pooled-energy test is therefore optimal there, and by the Neyman–Pearson lemma the count detector, whose false-alarm probability is at most the same 10^{-3}, cannot have a higher detection probability at any M. A violation would mean an error in Eq. (4) or (5).

5 Monte Carlo checks and the z-score

Verification 1 generates pilots, adds noise, subtracts the pilots and compares the observed P_{fa} and P_{d} with Eq. (4). Verification 2 simulates whole windows and compares $P(\text{count} \geq k)$ with Eq. (5), with and without the jammer. Each point is tested by

$$z = \frac{K - Np}{\sqrt{Np(1-p)}}, \tag{7}$$

where K is the number of events in N trials. S0 used $\sqrt{Np}$ in the denominator, the small-p form; here p can be close to 1, so the full binomial variance is needed. The pass criterion is $|z| < 4$ at every point, as in S0 and S1.

6 What the numbers show

With $n = 16$ pilots (`results/s3a_summary.txt`):

- At the S1 outage point and the jammer's worst-case $\rho = 0.0095$, per-hop detection is almost certain ($a = 12.7$), so $M^* = 242$ equals the strong-jammer bound (6): the delay is set by how rarely a hop lands in the jammed band. More pilots do not help ($M^* = 242$ for $n = 16$ and 32, 261 for $n = 8$). If the jammer must occupy at least one channel ($\rho = 0.01$), $M^* = 230$.
- At the same J/S with $\rho = 1$, each hop is only weakly affected ($a = 1.11$). The count detector needs 337 hops and the optimal pooled-energy test 212; here more pilots help roughly in proportion (671, 337, 170 hops for $n = 8, 16, 32$).
- Between the two, the count detector needs the fewest hops (92 on the grid, near $\rho \approx 0.04$). This is an observation about this detector only (Section 3), not a statement about the best possible detector.
- At $J/S = 12\,\text{dB}$, the S1 worst-case $\rho = 0.135$ needs 16 hops, again the bound (6); at $\rho = 1$ it needs 4 (pooled energy 3).

All M^* are in hops, not seconds: converting them needs a hop rate, which belongs to a specific radio and is not modelled here.

7 Correspondence with the code

Mathematics	Code
a, Eq. (2)	`noise_ratio()`
T, Eq. (1)	`simulate_hop_energy()`
γ, P_d from P_fa, Eq. (4)	`hop_detection()`
smallest admissible k	`count_threshold()`
M^* over the threshold grid	`hops_to_detect()`
pooled energy at $\rho = 1$	`pooled_detection()`, `hops_to_detect_pooled()`
Eq. (6)	`strong_jammer_limit()`
S1 outage point and worst-case ρ	imported from `s1_fh_partial_band_jamming.py`

References

[1] H. Urkowitz, "Energy detection of unknown deterministic signals," *Proceedings of the IEEE*, vol. 55, no. 4, pp. 523–531, Apr. 1967.